Solutions of Tutorial 6
ME101 Engineering Mechanics
Virtual Work, Stability, and Moment of Inertia

Q1. Predict through calculation whether the homogeneous semi-cylinder and the half-cylindrical shell will remain in the positions shown in Figure 1 or whether they will roll off the lower cylinders (i.e., whether there is a stable equilibrium or an unstable equilibrium for the two conditions).

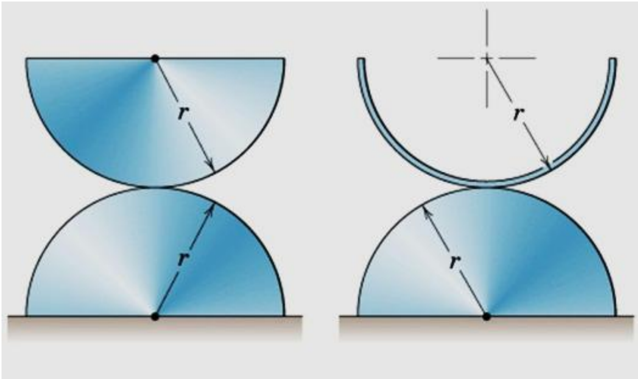


Figure 1

Solution of Q1: Using Potential Energy approach:

Let m be the mass of the top body

C.G. of semi-circular area $\bar{r} = \frac{4r}{3\pi}$ 1 mark

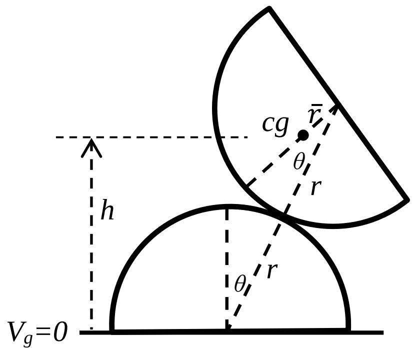
C.G. of semi-circular arc $\bar{r} = \frac{2r}{\pi}$ 1 mark

Considering datum at the base of the bottom cylinder.

Total potential energy of the system:

$V = V_g = mgh$ 1 mark

h is the distance between cg and the datum



1 mark

$V_g = mg\,(2r\cos s\theta - \bar{r}\cos 2\theta)$ 1 mark

$$\frac{dV}{d\theta} = mg(-2r\sin\theta + 2\bar{r}\sin 2\theta)$$ 1 mark

$$\frac{d^2V}{d\theta^2} = 2mg(-r\cos\theta + 2\bar{r}\cos 2\theta)$$ 1 mark

Checking equilibrium of the body at $\theta = 0$

For $\theta = 0$, $\frac{d^2V}{d\theta^2} = 2mrg(-1 + \frac{2\bar{r}}{r})$ 1 mark

For Cylinder: $\frac{\bar{r}}{r} = \frac{4}{3\pi}$, $\frac{d^2V}{d\theta^2} = 2mrg(-1 + \frac{8}{3\pi})$ = (negative) → Unstable Equilibrium 1 mark

For Shell: $\frac{\bar{r}}{r} = \frac{2}{\pi}$, $\frac{d^2V}{d\theta^2} = 2mrg\left(-1 + \frac{4}{\pi}\right)$ = (positive) → Stable Equilibrium 1 mark

Q2. Determine the area moment of inertia of the shaded area (Figure 2) I_{xy}.

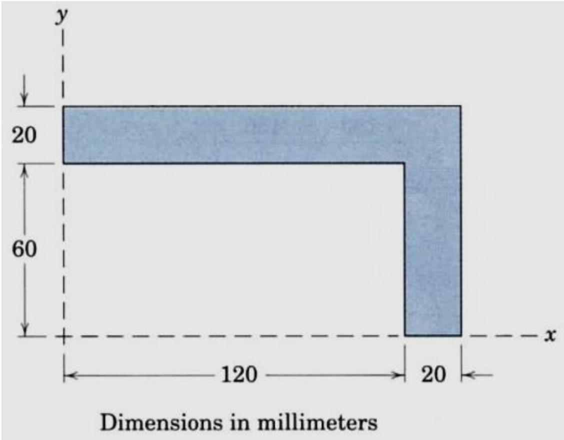
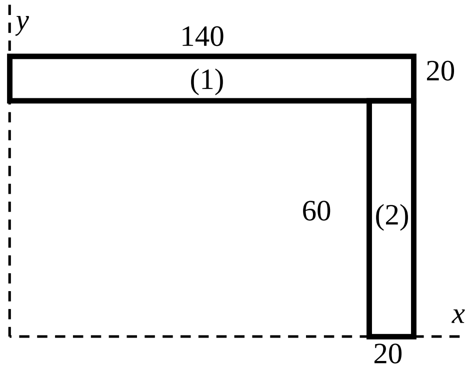


Figure 2



Solution of Q2:

Parallel axis theorem: $I_{xy} = \bar{I}_{xy} + d_x\, d_y\, A$ — 2 mark

Both areas (1) and (2) are symmetric about their respective centroidal Axis — 2 marks

→ $\bar{I}_{xy} = 0$ for both area.

Therefore, for Area (1): $I_{xy1} = d_{x1}\, d_{y1}\, A_1$

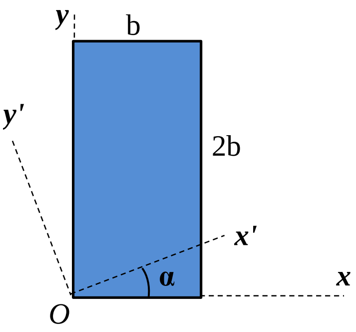
$I_{xy1} = 70\times70\times20\times140 = 13.72\times10^6$ mm ^{4} — 2 marks

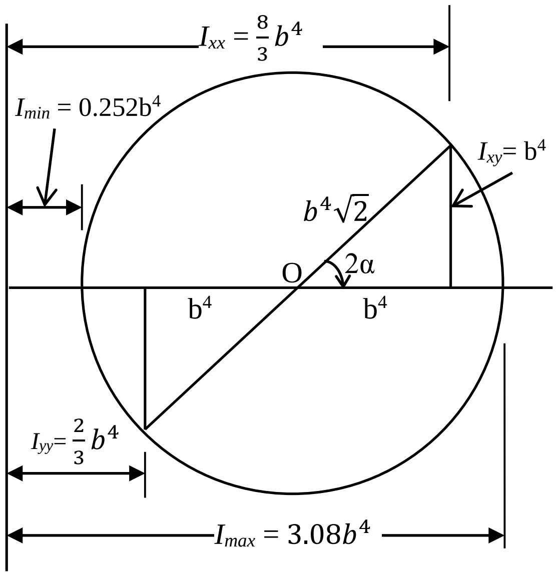
Similarly, for Area (2): $I_{xy2} = d_{x2}\, d_{y2}\, A_2$

$I_{xy2} = 130\times30\times60\times20 = 4.68\times10^6$ mm ^{4} — 2 marks

Total $I_{xy} = I_{xy1} + I_{xy2} = 18.40\times10^6$ mm ^{4} — 2 marks

Q3. Determine the angle α which locates the principal axes of inertia through point O for the rectangular area (Figure 3). Construct the Mohr's circle of inertia. Specify the corresponding values of I_{max} and I_{min}.





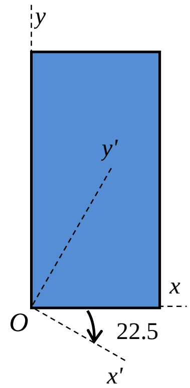


Figure 3

Solution of Q3:

$I_x = \frac{1}{3}b(2b)^3 = \frac{8}{3}b^4$ 1 mark

$I_y = \frac{1}{3}2b(b)^3 = \frac{2}{3}b^4$ 1 mark

$I_{xy} = 2b^2(\frac{b}{2})(b) = b^4$ 1 mark

1 mark

With this data, plot the Mohr's Circle, and using trigonometry calculate the angle 2 α

3 marks

2 α =$tan^{-1}(b^4/b^4)$ = 45°

1 mark

Therefore, α =22.5° (clockwise w.r.t. x)

This is equivalent to calculating:

$$\tan 2\alpha = \frac{2I_{xy}}{I_y - I_x} = \frac{2b^4}{\frac{2}{3}b^4 - \frac{8}{4}b^4} = -1$$

→ 2α = -45°, α = -22.5°

From Mohr's Circle:

$I_{max} = \frac{5}{3}b^4 + b^4\sqrt{2} = 3.08b^4$ 1 mark

$I_{min} = \frac{5}{3}b^4 - b^4\sqrt{2} = 0.252b^4$ 1 mark

Q4. If the parabolic profile in Figure 4 follows $y = a + bx^2$ then

a) Determine value of *a* and *b* [2 marks]

b) Determine total shaded area [3 marks]

c) Determine total area moment of inertia about y-axis of the shaded area [3 marks]

d) Determine the radius of gyration about the y-axis of the shaded area [2 marks]

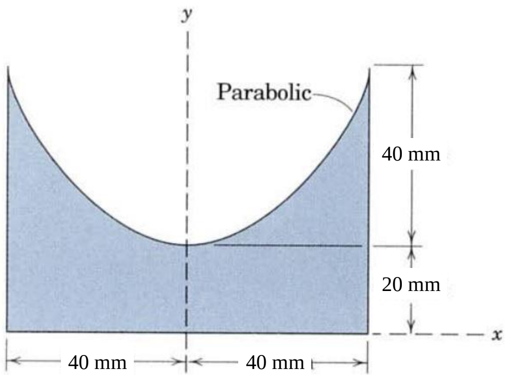


Figure 4

Solution of Q4:

Step 1: Determine value of *a* and *b* [2 marks]

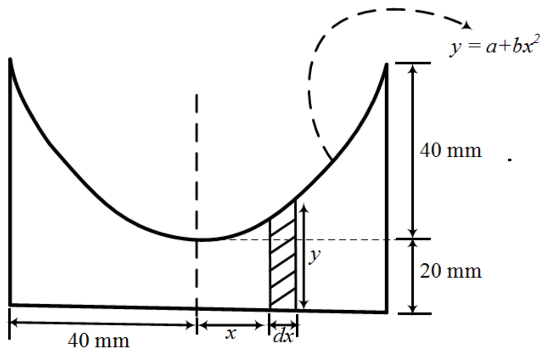


Figure 4.1

Equation of parabola $y = a + bx^2$

From figure 4.1, When *x* = 0 then *y* = 20 mm

$$20 = a + b(0^2)$$

$$a = 20 \text{ mm}$$

When *x* = 40 mm then *y* = 60 mm

$$60 = a + b(40^2)$$

$$60 = 20 + b(40^2)$$

$$b = \frac{1}{40 \text{ mm}}$$

Step 2: Determine total shaded area [3 marks]

Elemental area dA

$$dA = y.dx$$

Total area $A = \int dA$

$$A = \int dA = \int y.dx = 2\int_0^{40} \left(20 + \frac{1}{40}x^2\right) dx$$

$$= 2\left[20x + \frac{x^3}{120}\right]_0^{40}$$

$$= 2666.67 \text{ mm}^2$$

Step 3: Determine total area moment of inertia about y-axis of the shaded area [3 marks]

Elemental area of moment about *y-axis*

$$dI_y = dA.x^2$$

Total moment of area about *y-axis*

$$I_y = \int dI_y = \int x^2 y dx = 2\int_0^{40} [20x^2 + \frac{x^4}{40}] dx$$

$$= 2\left[\frac{20}{3}x^3 + \frac{x^5}{200}\right]_0^{40} = 1.88 \times 10^6 \text{ mm}^4$$

Step 4: Determine Radius of gyration about y-axis [2 marks]

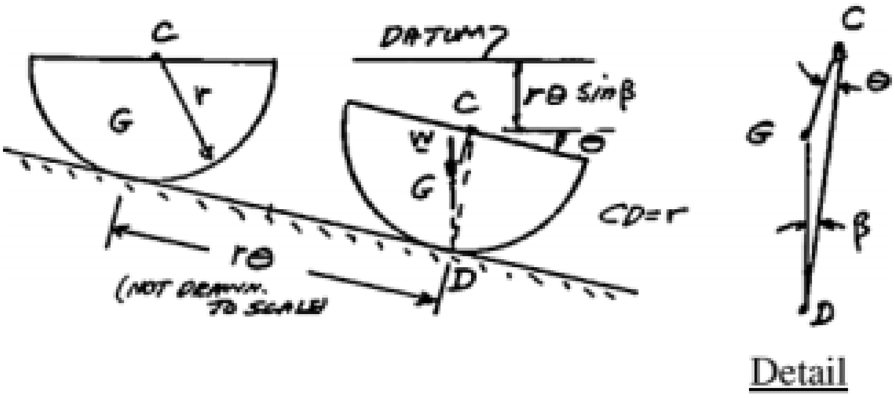
Radius of gyration about *y-axis* = $\sqrt{\frac{\text{Total moment of area about } y\text{−}axis}{\text{Total area}}}$

$$k_y = \sqrt{\frac{I_y}{A}} = \sqrt{\frac{1.88\times10^6}{2666.67}} = 26.55 \text{ mm}$$

Q5. A homogeneous hemisphere of radius **r** is placed on an inclined plane as shown **Figure 5**. Assuming that the friction is sufficient to prevent slipping between the hemisphere and the inclined plane, determine

a) The largest angle β for which a position of equilibrium exists. [**6 Marks**]

b) The angle θ corresponding to equilibrium, when the angle β is equal to half the value found in part **a.** [**4 Marks**]



$$CG = \frac{3}{8}r$$

$$V = w(-r\theta sin\beta - (CG)cos\theta)$$

$$\frac{dV}{d\theta} = w\left(-rsin\beta + \frac{3}{8}rsin\theta\right)$$

$$\frac{dV}{d\theta} = 0;$$

$$-rsin\beta + \frac{3}{8}rsin\theta = 0$$

$$sin\beta = \frac{3}{8}sin\theta \qquad (1)$$

For $\boldsymbol{\beta_{max}}$, $\boldsymbol{\theta = 90^0}$

From (1) $sin\beta = \frac{3}{8}sin90$

$\boldsymbol{\beta_{max} = 22.02^0}$ **[6 marks]**

If $\boldsymbol{\beta = 0.5\beta_{max} = 11.01^0}$

From (1), $sin11.01 = \frac{3}{8}sin\theta \Rightarrow \boldsymbol{\theta = 30^0}$ **[4 marks]**